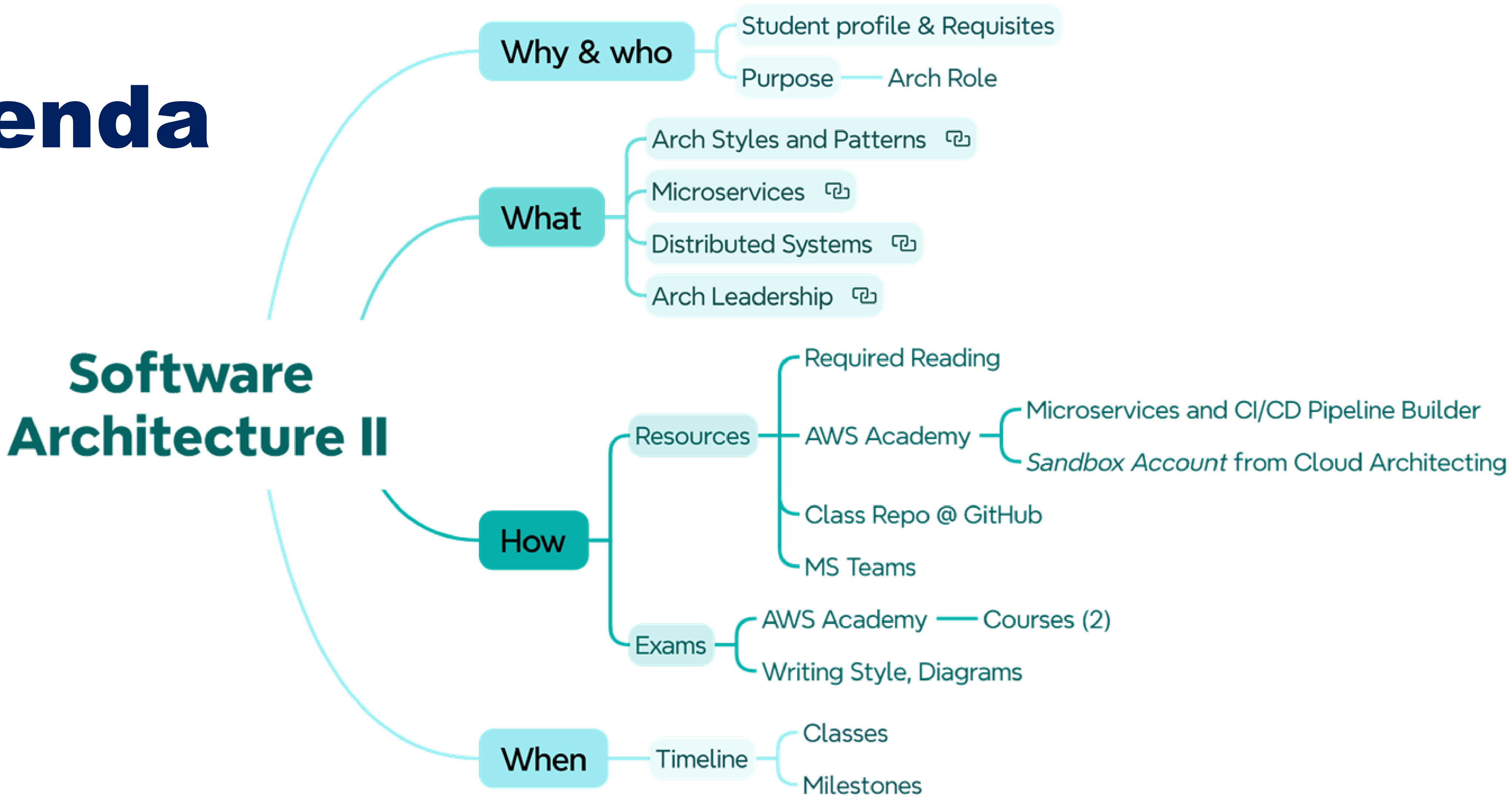
Software Architecture II

Arquitectura de Software II

Universidad de La Sabana

FACULTAD DE INGENIERÍA

Agenda


Software Architecture II
Why & who
Student profile & Requisites
Purpose
Arch Role
What
Arch Styles and Patterns
Microservices
Distributed Systems
Arch Leadership
How
Resources
Required Reading
AWS Academy
Microservices and CI/CD Pipeline Builder
Sandbox Account from Cloud Architecting
Class Repo @ GitHub
MS Teams
Exams
AWS Academy
Courses (2)
Writing Style, Diagrams
When
Timeline
Classes
Milestones

Why: Student Profile

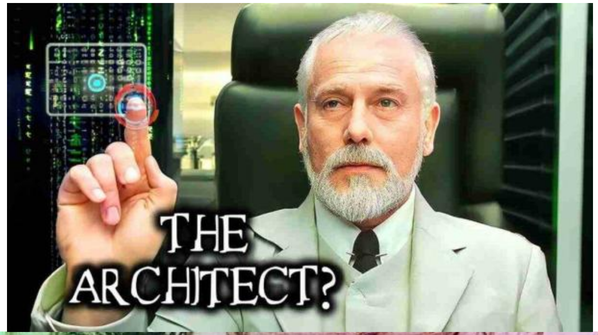
- Curious and Discipline: Read IT topics on several reputable sources and plan his/her works.
- Dev: AWS Labs use Node, but knowledge of a modern language is enough (Go, Rust, Clojure, Java, Python).
- Usage of GenAI Tools for specific tasks only.
- Attendance to avoid repeat or misunderstanding

Prerequisites:

- Containerization: Usage of Docker CLI. Course on this MSc (Arquitectura de Software I).
- Database: Knowledge on NoSQL DB Design. Course on this MSc (Diseño y Optimización de Bases de Datos).
- Cloud: Knowledge on AWS. Course on this MSc (Arquitecturas en la Nube). Focus on ECS, CodePipeline and CodeDeploy (Asynch Source-To Review).

Universidad de La Sabana
FACULTAD DE INGENIERÍA

Why: Architecture Role



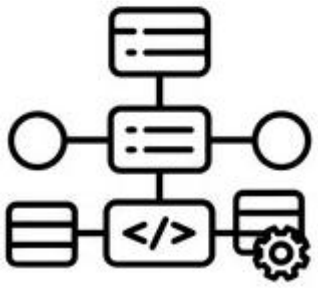
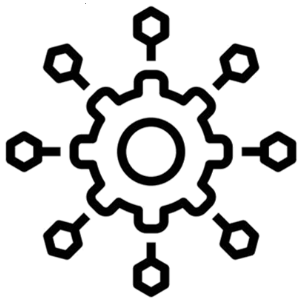
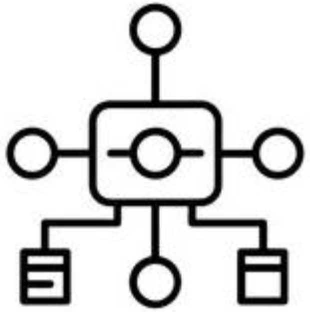
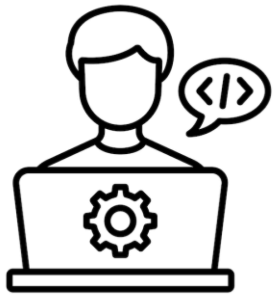
- To set responsibilities of an architect role.
- To apply proven (industry) methods to solve problems.
- To learn how to make good things (best practices), possible things (constraints) and control exceptions (antipatterns).

See laws and expectations before *Required Reading*

My experience, reputable authors and industry

Relevant Significance, unknown architects

What

Estilos y Patrones Arquitectonicos	Architectural Styles and Patterns	
Microservicios	Microservices	
Sistemas Distribuidos	Distributed Systems {Platforms, Infrastructure, Computing}	
Liderazgo en Arquitectura	Architecture Leadership	

How: Resources

01 **AWS Academy**

02 **Class Repo@GitHub**

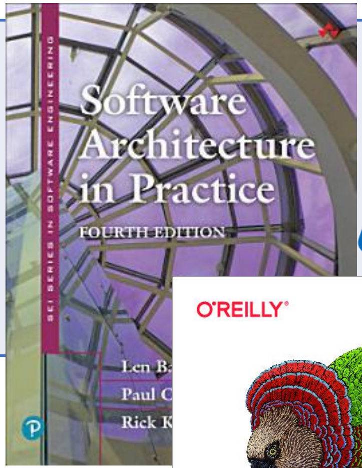
03 **MS Teams**

04 **Unisabana e-Learning**

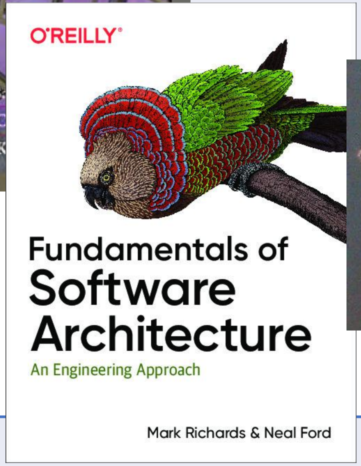
Note: Take advantage of AWS Academy promos. All links or books are referenced on the slide or at the end of presentation.

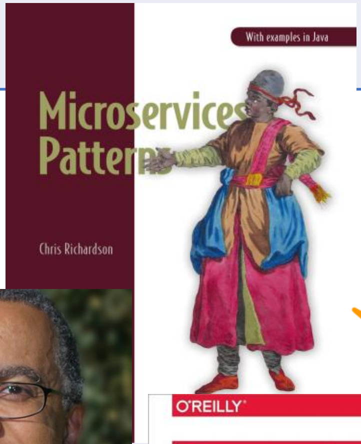
How: Required Reading

Architectural Styles and Patterns



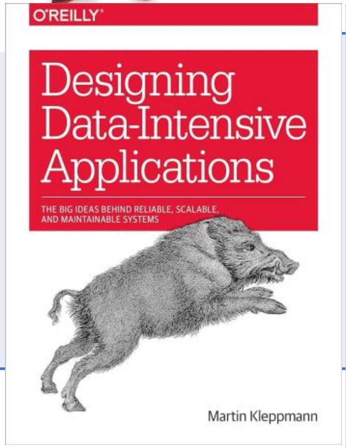
Microservices

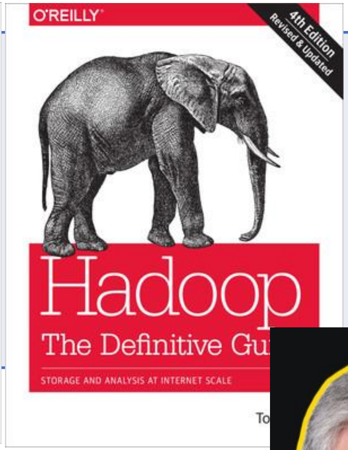




Distributed Systems {Platforms, Infrastructure, Computing}

Base





Architecture Leadership

[Many......]

See How my reference system at the end

How: Exams

Both Groups: Tuesday and Wednesday

Item	%	Deadlines	Type	On
Arch Styles and Patterns	20	5th Class (9 Aug)	Doc (Upto 2)	Uni e-learning
Microservices	50	5th Class (9 Aug)	Doc & Lab	AWS Academy & Uni e-Learning
Distributed Systems	15	Next Class (23 Aug)	Doc & Lab	AWS Academy & Uni e-Learning
Arch Leadership	15	Last Class (2 Sept)	Docs (Upto 2)	Uni e-learning

Time to run scripts for labs (AWS Academy only): Saturdays, Sundays and Holidays at 5 pm. You let the credentials before of that hour. Exceptional cases can be discuss on this class only.

AWS Academy issues: Create support the ticket directly and keep inform the professor. AWS Academy will need all screenshots.

Unisabana e-Learning issues: Review with University Point-of-Contact, the IT support or Carol Daniela Suarez Mendez (TA).

Extra Time with regular process **only**: Review with University Point-of-Contact or TA, the procedure and evidences.

Activity Questions: Leave the questions on “Foro de dudas e inquietudes U1” **only**. The professor will subscribe to that forum **only**.

Take care of the end date for Courses on AWS Academy to take advantage of the promos and labs.

AWS Academy

- Task 2: For this course, Microservices and CI/CD Pipeline Builder.
- Task 3: Take advantage of the Sandbox Account from previous course, “Cloud Architecting”.

Those tasks are evaluated by the professor using scripts. AWS Academy doesn’t offer a grader for those elements.
Your responsability are to have the resources running on your account, the professor run scripts as well you do.

Preferred style

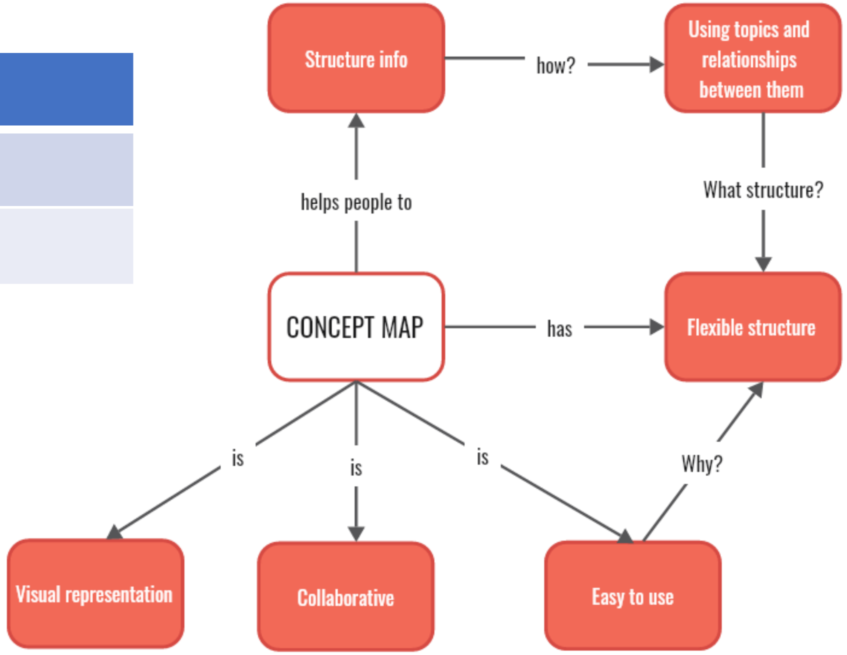
Simple

Well-Structured

Relevant info

Header1	Header2
Or using	A table
With clever	order

- Using bullets,
- It can be a
- Great idea!



When: Timeline

https://unisabanae-learning.unisabana.edu.co/course/view.php?id=26430

Class #		Module
1	14,15 Jul	0, 1
2	21, 22 Jul	1
3	28, 29 Jul	2
4	11, 12 Aug	2 + SDD
5	18, 19 Aug	3
6	25, 26 Aug	4 + Interviews

I added the Classroom Presentation (PPT) on the Github on the same week.

Review the grade criteria on each unit for “Actividades de aprendizaje”, “Guia de Aprendizaje”.
Last time for questions for each unit are on the previous week of the deadline.

Universidad de La Sabana
FACULTAD DE INGENIERÍA

Why I started with.....

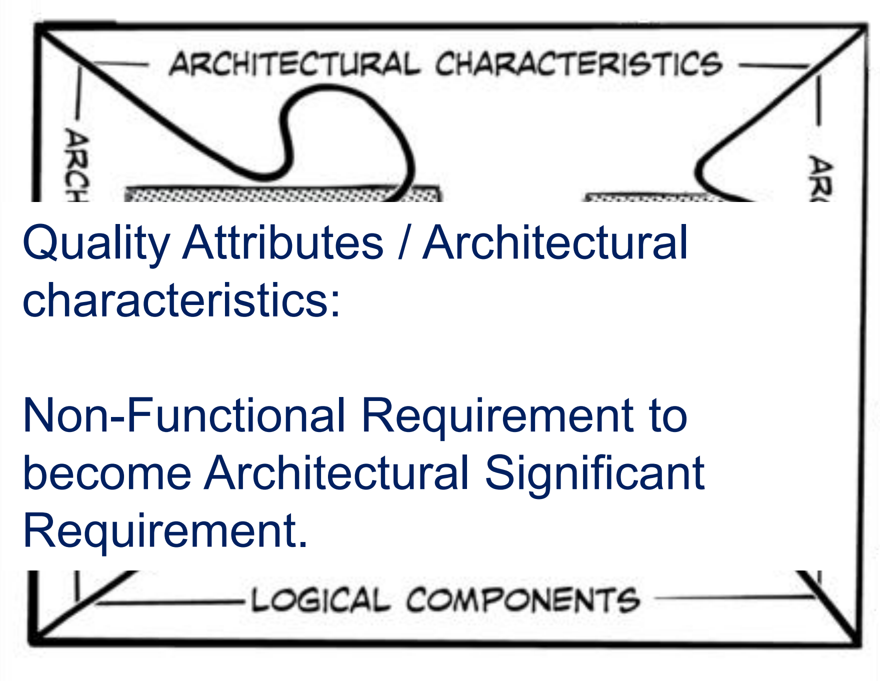
Separation of concerns → Technical (Previous Course).

Started with "Why" → Purpose

Deep Work → Focus on limited time

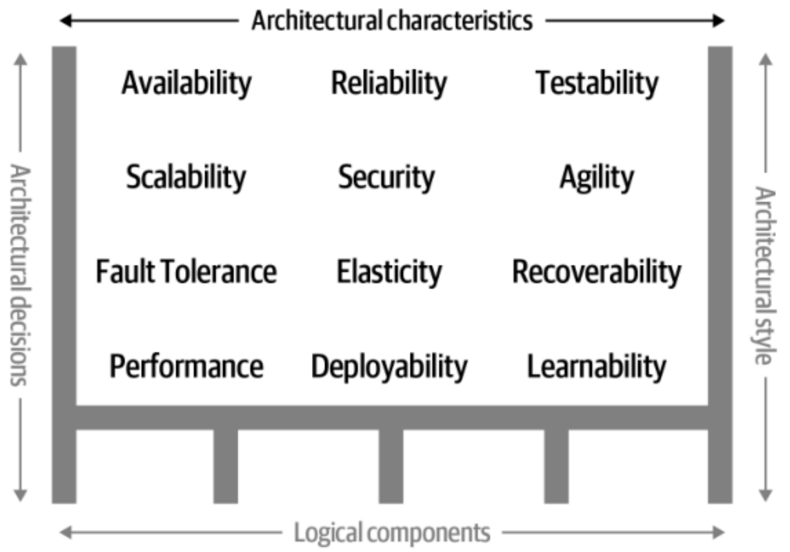
Simplicity → Master

SA Dimensions: Quality Attributes

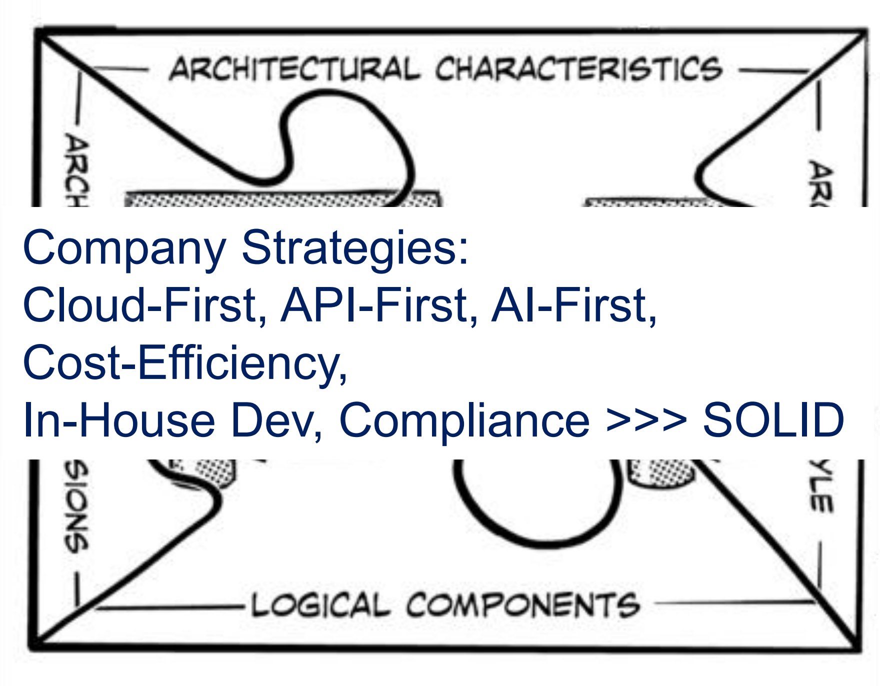


Quality Attributes / Architectural characteristics:

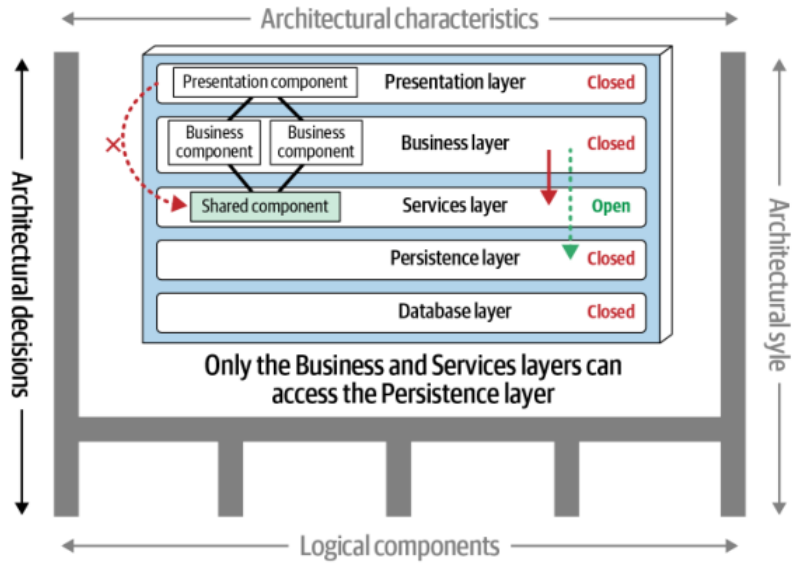
Non-Functional Requirement to become Architectural Significant Requirement.



SA Dimensions: Architectural Decisions

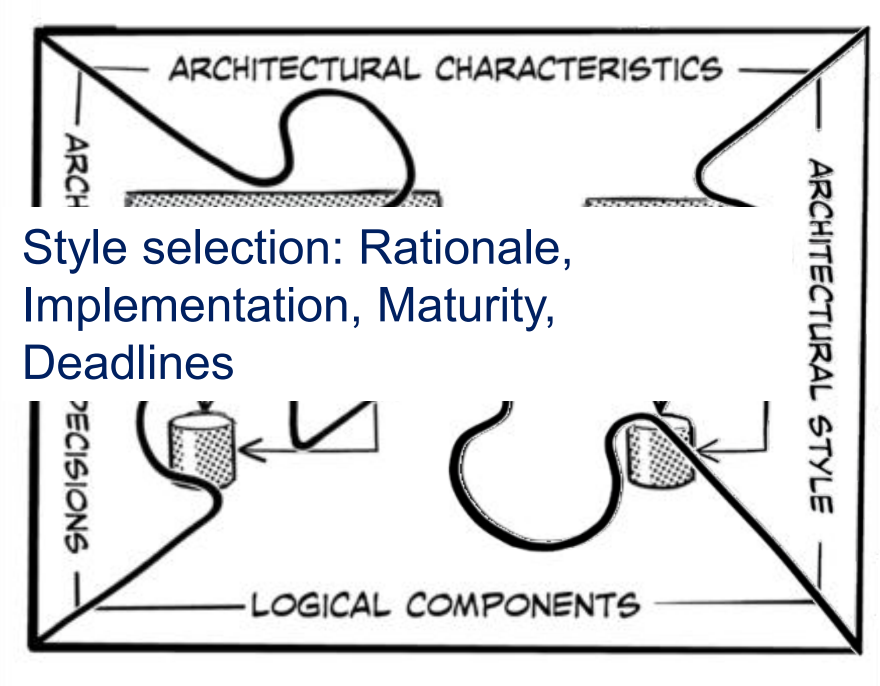


Company Strategies:
Cloud-First, API-First, AI-First,
Cost-Efficiency,
In-House Dev, Compliance >>> SOLID

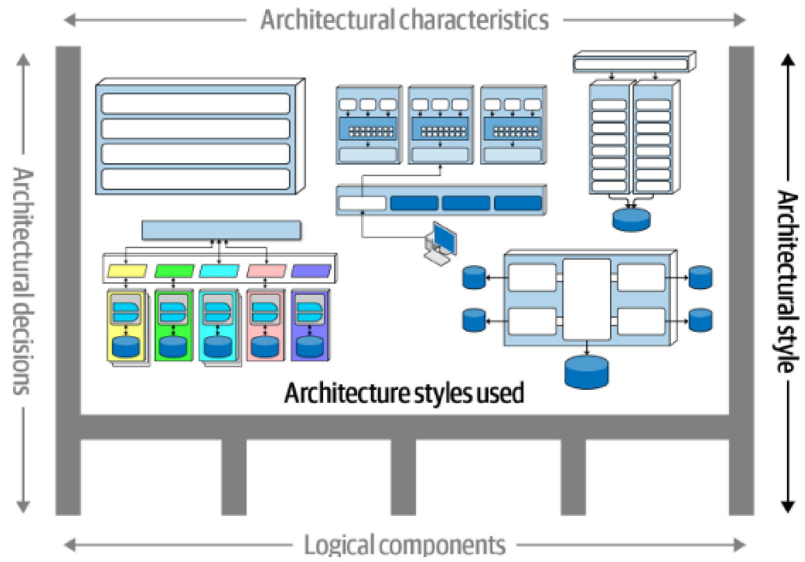


Universidad de La Sabana
FACULTAD DE INGENIERÍA

SA Dimensions: Architectural Styles

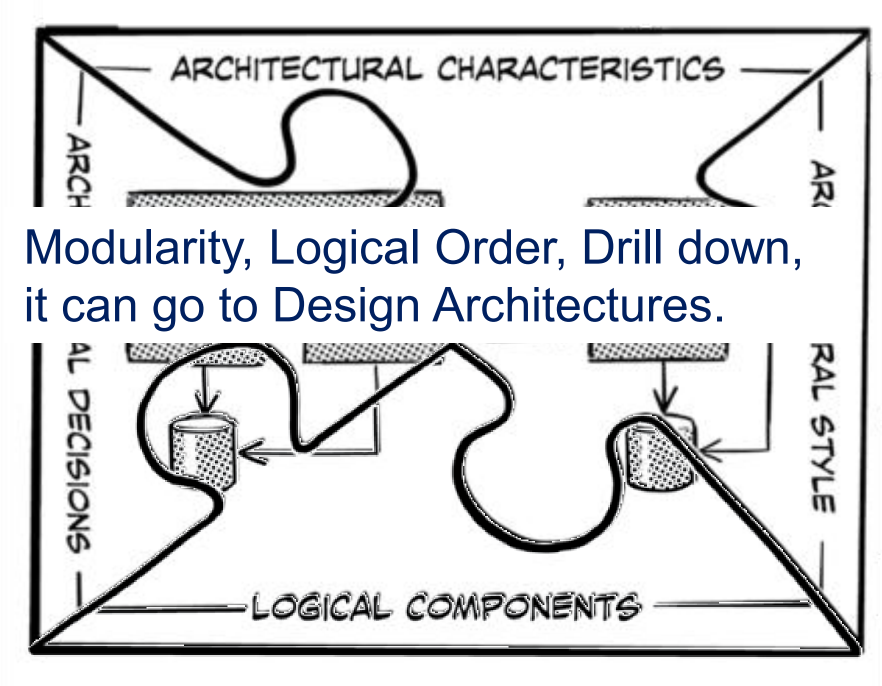


Style selection: Rationale, Implementation, Maturity, Deadlines

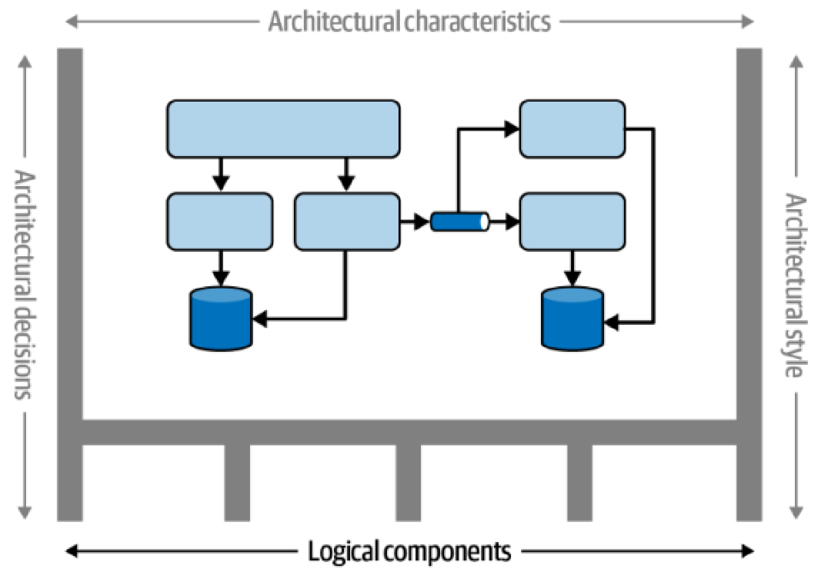


[FoSA] and [HFSoA]

SA Dimensions: Logical Components



Modularity, Logical Order, Drill down, it can go to Design Architectures.



Universidad de La Sabana
FACULTAD DE INGENIERÍA

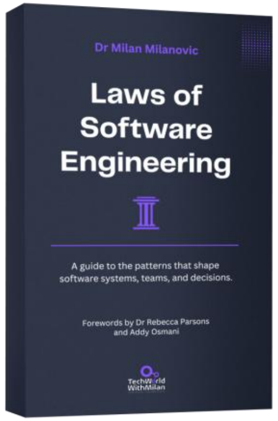
Scope and Laws

Leadership

- Transparent
- Discipline
- Original

Laws of Software Architecture

1. Everything in software architecture is a trade-off.
2. Why is more important than how.
3. Most architecture decisions aren´t binary but rather exist on a spectrum between extremes.



Conway's law, YAGNI, CAP Theorem, Fallacies of Distributed Computing.

Simplicity (3), Cognitive Bias (4).

Expectations

- Make architecture decisions
- Continually analyze the architecture
- Ensure compliance with decisions
- Keep current with latest trends
- Understand diverse technologies, frameworks, platforms, and environments
- Know the business domain
- Lead a team and possess interpersonal skills
- Understand and navigate organizational politics

[FoSA]

Good Quotes

Architecture is the stuff you can't Google or ask an LLM about.
—Mark Richards

Common sense? AI Impact?

There are no right or wrong answers in architecture—only trade-offs.
—Neal Ford

No silver bullet. Hold your ground and/or pick your battles.

Programmers know the benefits of everything and the trade-offs of nothing. Architects need to understand both.
—Rich Hickey, Clojure Creator

Leadership and allies (mutual benefits)

Living Architectures >> Evolutionary Architectures

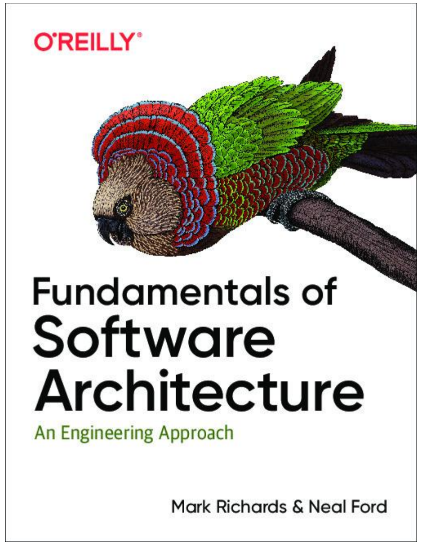
Universidad de La Sabana
FACULTAD DE INGENIERÍA

References

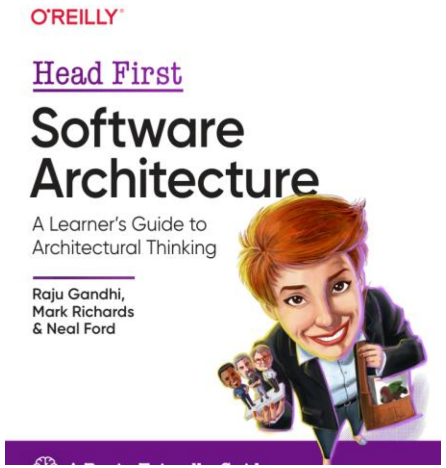
Links are everywhere can be exasperating, so it is better to have an ending slide with the chapters and the acronym.

The abbreviations to be referred are here.

- [FoSA] Richards, M., & Ford, N. (2025). Fundamentals of software architecture (2nd ed.). O’Reilly Media.
  Ch. 1, 26 & 27



- [HFSoA] Raju, S., Richards, M., & Ford, N. . (2024). Head First software architecture: A learner’s guide to architectural thinking (1.ª ed.). O'Reilly Media.
  Ch. 1, 2 & 3



[FoSA] and [HFSoA]

Universidad de La Sabana

FACULTAD DE INGENIERÍA

CONTACTO

Francisco Moreno

Profesor Catedra – Facultad de Ingeniería

franciscomodi@unisabana.edu.co